

Course: Networked Control

Lecture III.1: Introduction to networked control systems

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Networked control: from Part II to Part III

What is this course about?

Issues arising in the **design and implementation** of
control systems
characterized by the presence of
components arranged in a networked
configuration

Networked control: from Part II to Part III

Issues

MIMO systems

Very complex and large-scale system under control

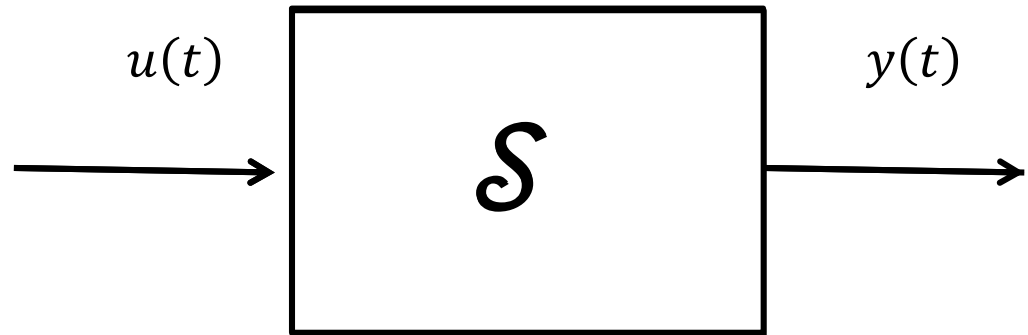
- Large number of interacting subsystems
- Complex system structure



High-dimensional and complex control problem.



- Need of «non centralized» control schemes
 - Subsystem/device level
 - Coordination between subsystems.



- Chemical plant
- Electric grid
- Fleet of autonomous underwater vehicles
- Irrigation network

Networked control: from Part II to Part III

Issues

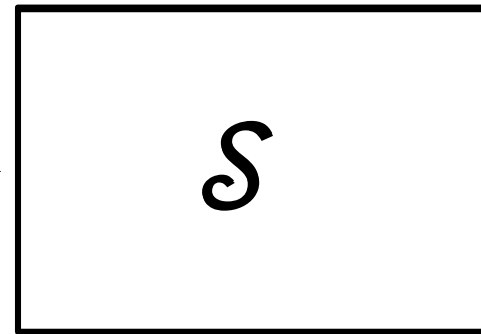
MIMO systems

Presence of a (possibly very) large number of sensors, actuators, and computing units, possibly in different locations and geographically distributed

Shared communication channel
(*control network*)

Communication issues

$u(t)$



$y(t)$

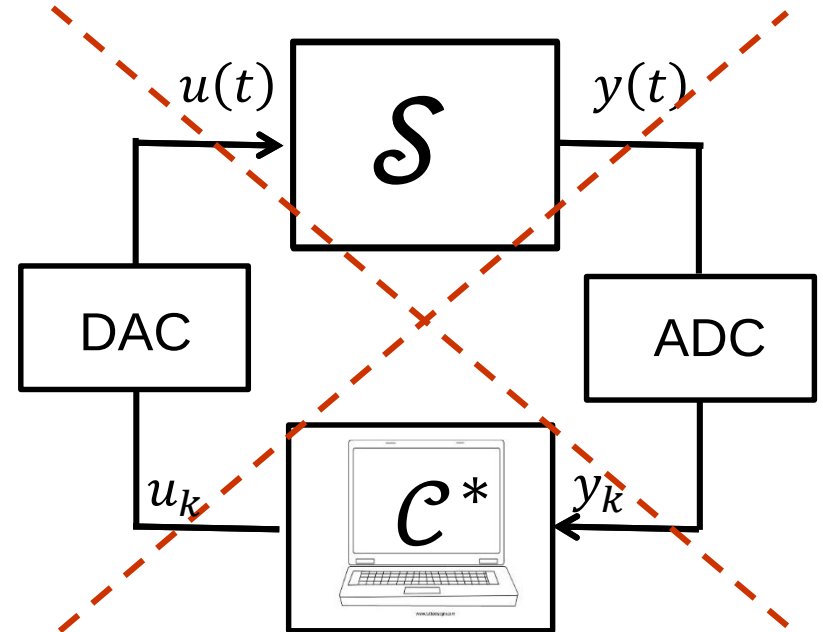
- Car
- Aircrafts
- ...

Networked control: from Part II to Part III

Issues

New control applications pose new **challenges**.

Classic **centralized** control solutions with **ideal communication** may be not feasible/possible/adequate



Networked control: from Part II to Part III

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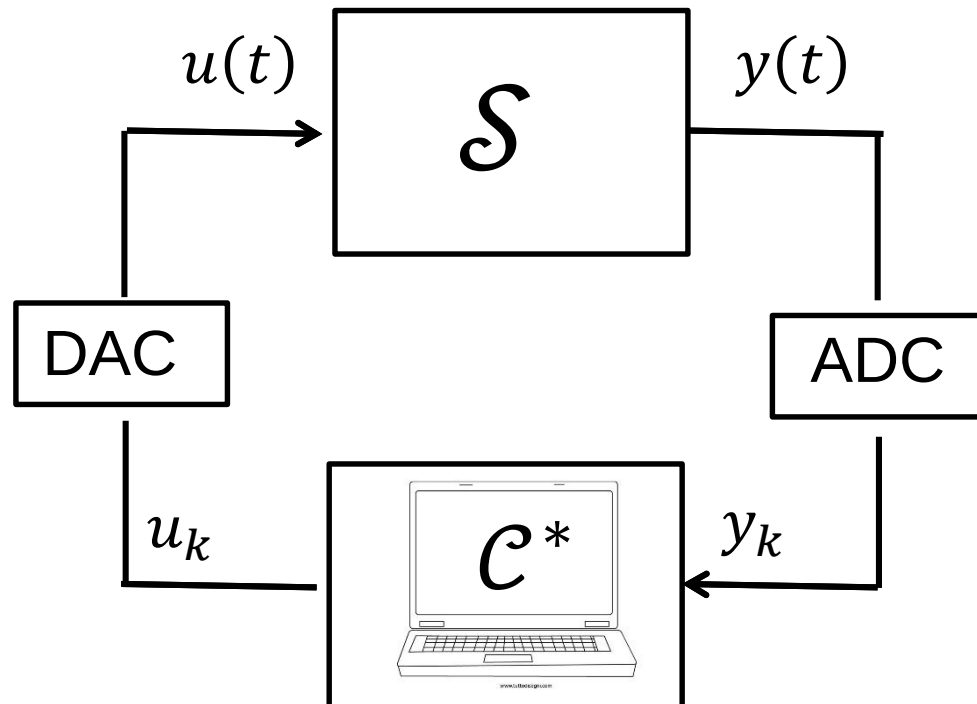
*Decentralized and
distributed control
(Part II).*



*Control under
non-ideal
communication
(Part III)*

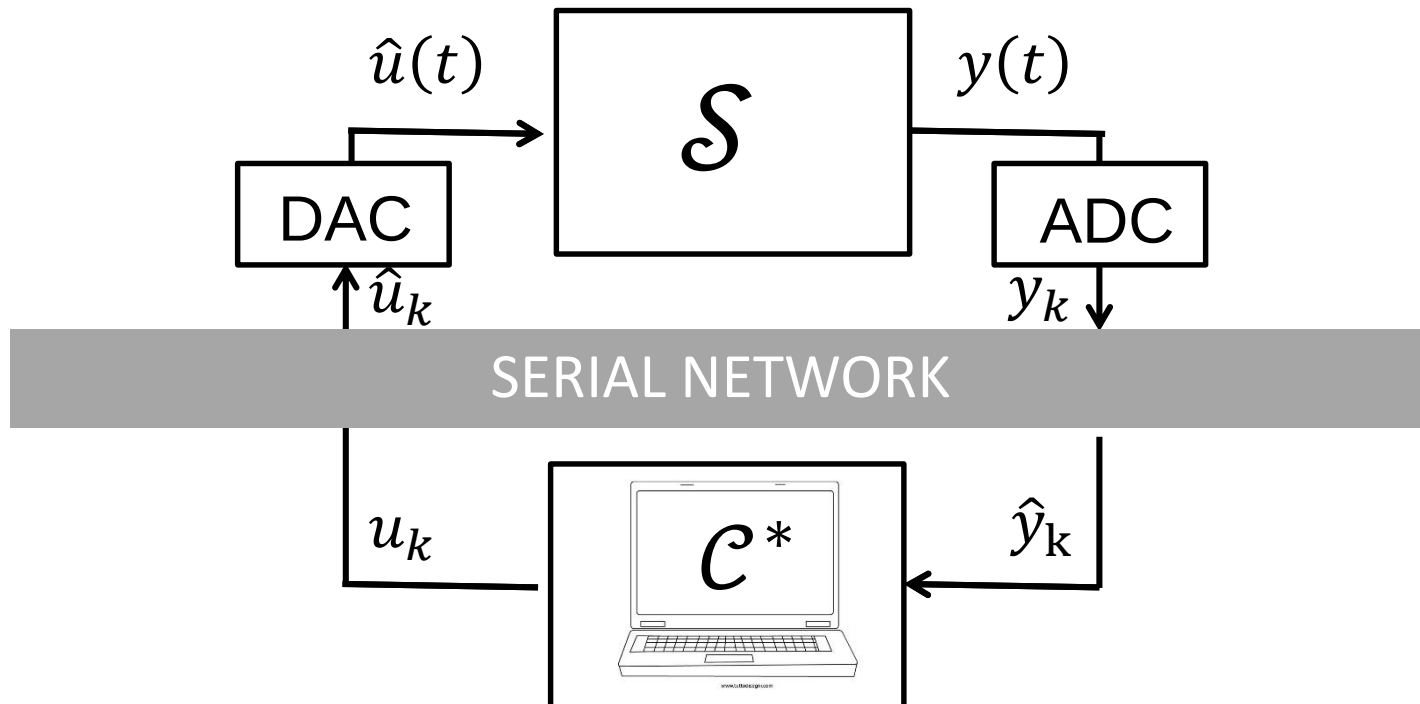
Networked control systems

- In the next part of this course we will deal with **control over shared serial communication channels**.

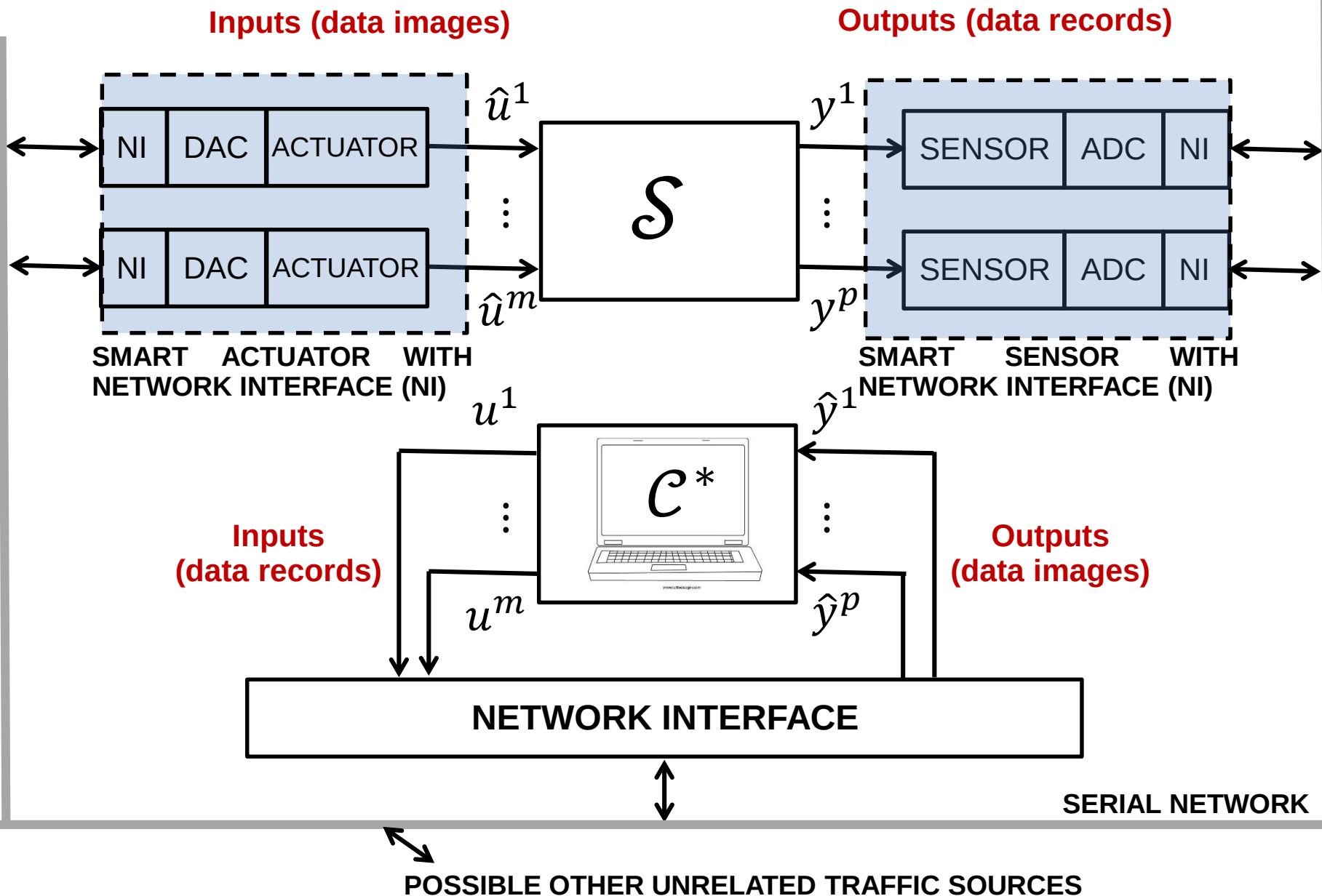


Networked control systems

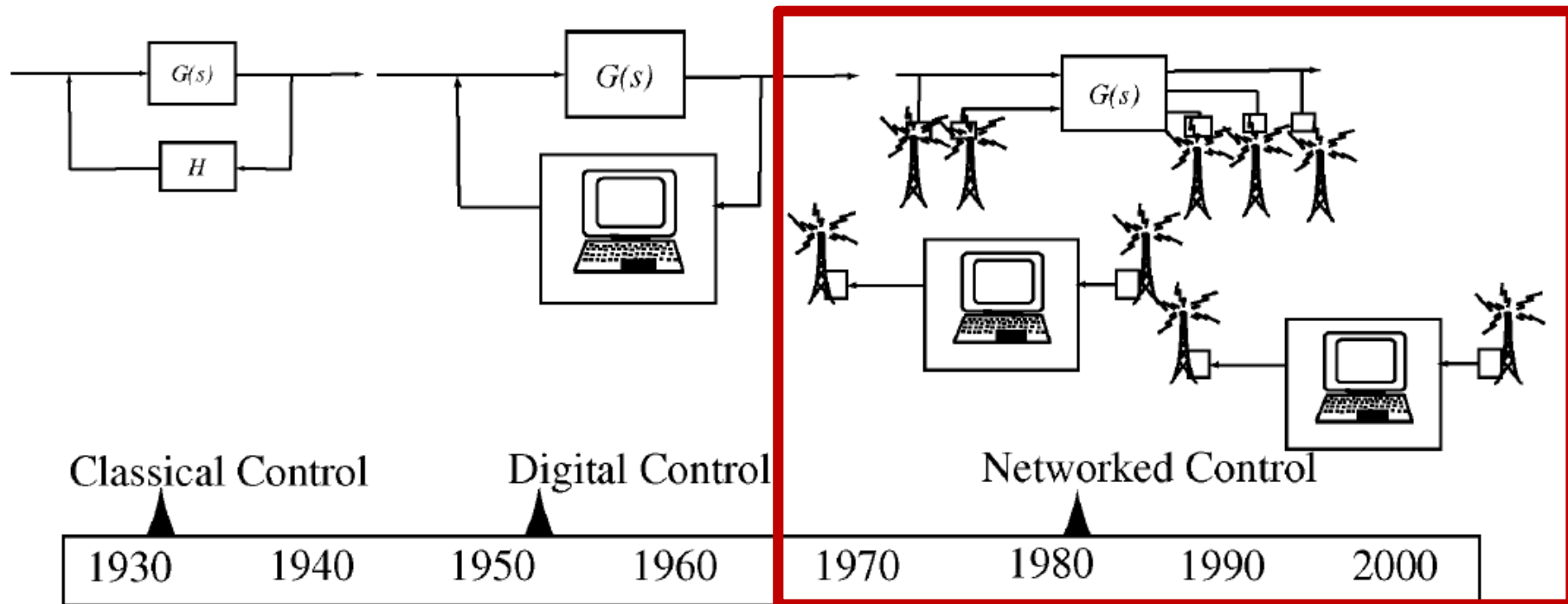
- In the next part of this course we will deal with **control over shared serial communication channels**.
- We define as a **networked control system** a **control system** whose parts (sensors, actuators, controllers) are connected through a serial communication network.
- A network is used, e.g., for communicating data
 - from the sensors to the controller, and/or
 - from the controller to the actuator



Networked control systems



Historical background



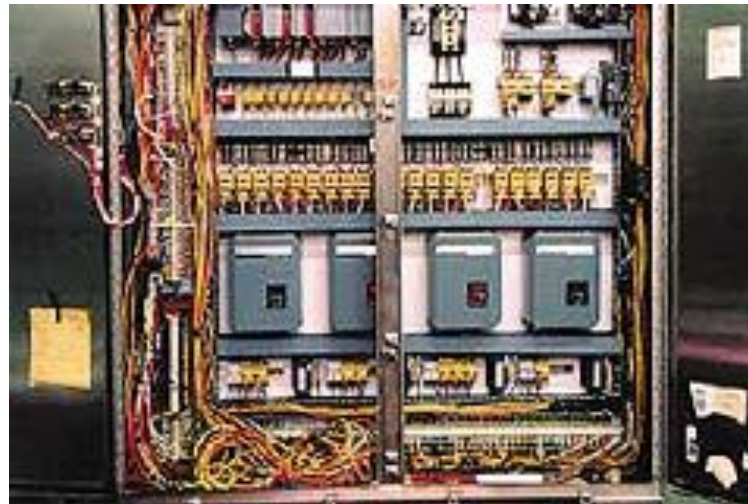
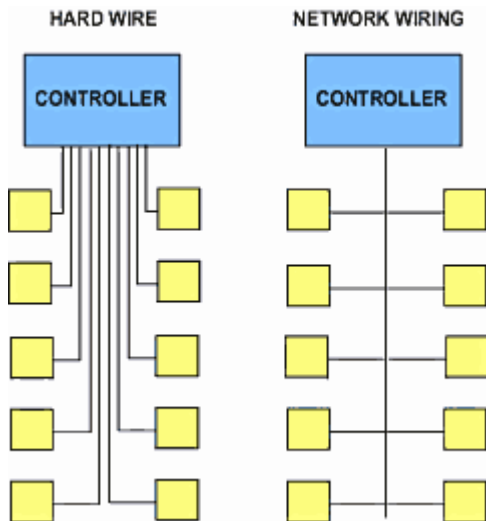
From: J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28

Historical background

- The beginning of research on **network control systems** can be traced back to 1983.
- In 1983 Bosch GmbH launched a feasibility study for the use of devices, connected using an *ad hoc* network, to control different functions in passenger cars. This was the earliest effort towards modern networked control.
- In 1986 the Control Area Network (CAN) protocol was announced at the Congress of the Society of Automotive Engineers, Detroit.
- In 1987 the CAN hardware (in the form of INTEL 82526 chip) was introduced.
- Today all cars manufactured in Europe embed systems connected through CAN.
- Networked control system technologies (with different protocols and network types) are nowadays fairly mature in a variety of industrial applications, e.g.,
 - Automobiles
 - Manufacturing plants/industrial automation
 - Aircrafts/spacecrafts
 - Heating, ventilation, and air conditioning (HVAC) systems
- Recent trends: integration through wireless rather than wired communication channels.

Benefits of networked control systems

- Often it is the **only technical solution**, e.g., when distances are large.
- **Cost**: save wiring when multiple devices are connected together.
 - A serial network consists of a single trunk line instead of hundreds of individual wires.
 - As the system grows past the deadline of 100 I/O points the added expense of network hardware is offset by savings in wiring time. This benefit grows exponentially as the size of the system.



Control panel with

- Point-to-point wired communication
- Devicenet

Courtesy from <https://www.perrymarshall.com/articles/industrial/part-1/>

Benefits of networked control systems

- **Modularity:** connectivity is achieved through software. It does not matter in which place of the network a component is connected.
 - Modularization of functionalities: decision and control functions can be distributed among different controllers in the network.
 - Availability of standard interfaces for interoperability and interchangeability.
 - Reconfigurability (simple installation and maintenance).
 - Plug and play functionalities.
 - Self configuration: controllers can detect which devices are connected and load suitable software configurations.
- **Diagnostics capabilities:** information (e.g., about malfunctionings of single devices) can be transmitted by the sensors and are made accessible through the network.
- Avoid single point-of-failure and **allow for redundant configurations.**

Challenges of networked control systems

(for wired, but especially wireless communication networks)

- **Band-limited channels:** constraint on the amount of information (number of packets and/or number of bits per packet) that can be transmitted per time unit.
 - Particularly critical in
 - Power-starved vehicles (e.g., planetary rovers, gliders).
 - Long endurance, energy-limited systems (sensor networks).
 - Micro-sensors and micro-actuators.
 - This has a major impact on the **tradeoff between sampling and quantization**.
- **Delays** (possibly time-varying), due to
 - Congestions.
 - Competition for communication resources, that depends on the network protocol, number of nodes, etc.
 - Transmission time, i.e., the time that a packet spends in the network.

Challenges of networked control systems

(for wired, but especially wireless communication networks)

- **Packet dropouts:** loss of packets during the transmission. Due to:
 - Errors in nodes and physical network links (more common in wireless than in wired networks).
 - Buffer overflow during congestions.
 - Long transmission delays or scheduling time. The receiver may discard the old data in real-time control applications.
- Noise.
- Fading, i.e., variation of the attenuation of a signal (with respect to, e.g., geographical position and radio frequency).

Challenges of networked control systems

Disclaimer:

In this course we will deal with

issues arising in control over network

i.e., impact of bandwidth limitations, delays, and packet dropouts on the stability properties of the **networked control system**.

We are **not interested in**

control of communication networks

e.g., maximization of the communication network throughput, development of efficient scheduling/routing algorithms.

Main related references

- G.C. Walsh, H. Ye, Scheduling of networked control systems. *IEEE Control System Magazine*, 21(1), 2001, pp. 57-65.
- R. J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28.
- Perry Sink Marshall, A comprehensive guide to industrial networks.
<https://www.perrymarshall.com/articles/industrial/part-1>